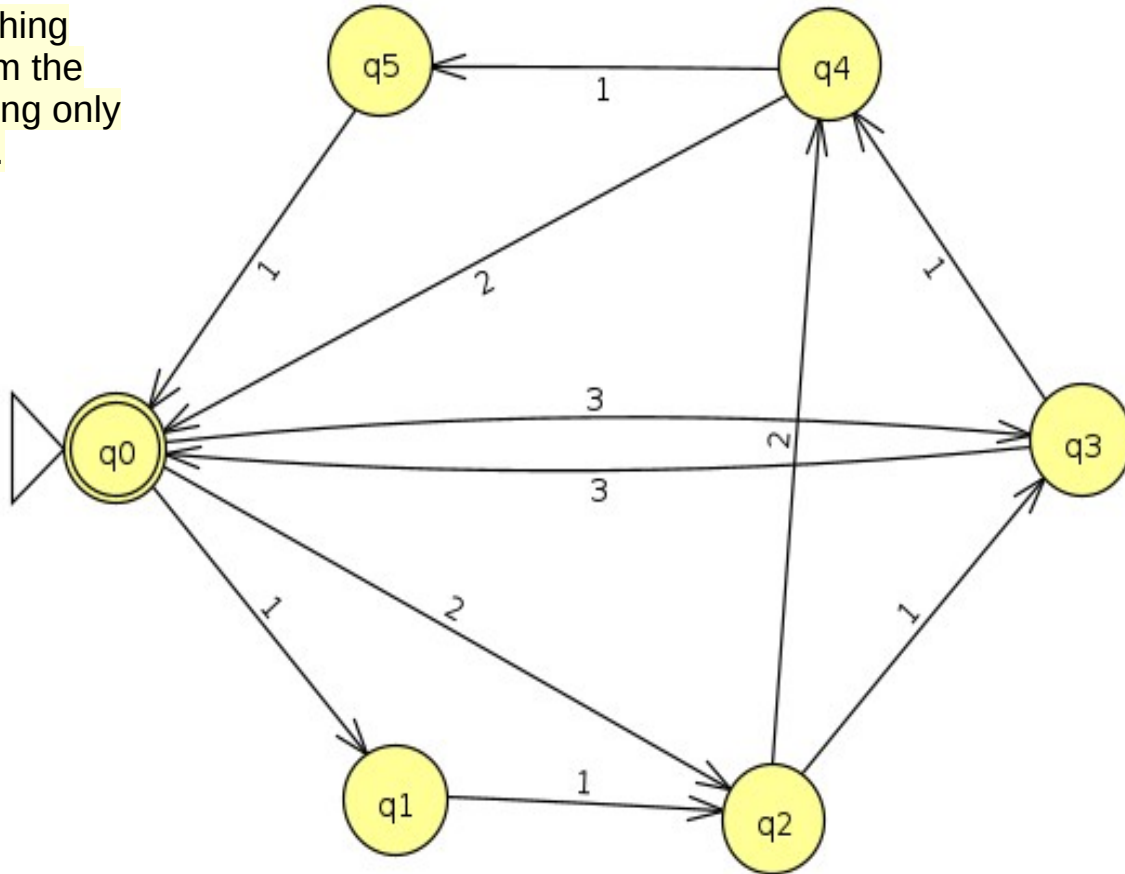


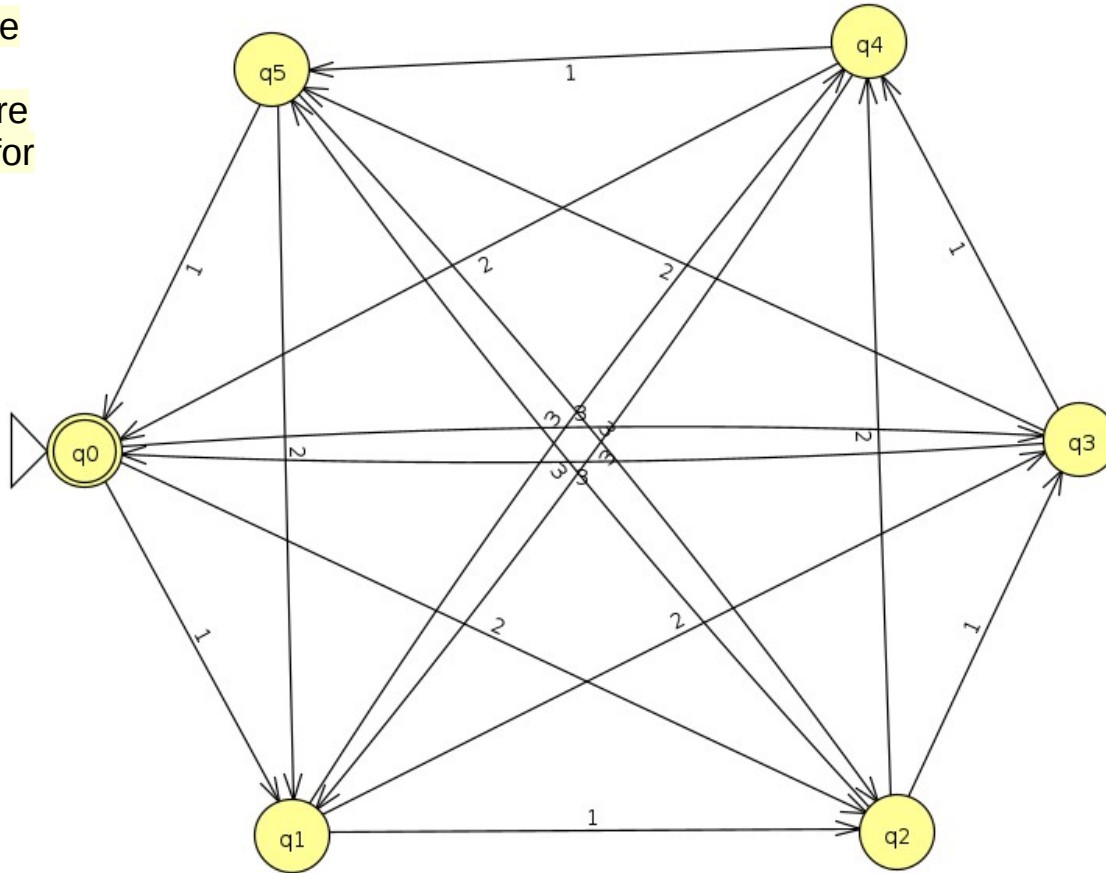
T1. Make an automaton: $\Sigma = \{1,2,3\}$ The automaton sums the numbers it receives modulo 6. The automaton accepts the input when it is 0 (mod 6).

1/2 First sketching transitions from the initial state using only input 1, 2 or 3.



T1. Make an automaton: $\Sigma = \{1,2,3\}$ The automaton sums the numbers it receives modulo 6. The automaton accepts the input when it is 0 (mod 6).

2/2 Then, make sure that from each state there is a transition for all three input symbols.



$Q = \{q_0, q_1, q_2, q_3, q_4, q_5\}$

$\Sigma = \{1, 2, 3\}$

$s = q_0$

$F = \{q_0\}$

$\delta = \{$

$\delta(q_0, 1) = q_1 \quad \delta(q_0, 2) = q_2 \quad \delta(q_0, 3) = q_3$

$\delta(q_1, 1) = q_2 \quad \delta(q_1, 2) = q_3 \quad \delta(q_1, 3) = q_4$

$\delta(q_2, 1) = q_3 \quad \delta(q_2, 2) = q_4 \quad \delta(q_2, 3) = q_5$

$\delta(q_3, 1) = q_4 \quad \delta(q_3, 2) = q_5 \quad \delta(q_3, 3) = q_0$

$\delta(q_4, 1) = q_5 \quad \delta(q_4, 2) = q_0 \quad \delta(q_4, 3) = q_1$

$\delta(q_5, 1) = q_0 \quad \delta(q_5, 2) = q_1 \quad \delta(q_5, 3) = q_2$

$\}$

T2. Modify the T1 automaton so that it also accepts negative values -1,-2,-3

Even if JFLAP accepts more than one input character to transition (so theoretically we could just add backward transitions with "-1", "-2", etc.), there really should be separate transition with - and digit. DFA transition function is $\delta: Q \times \Sigma \rightarrow Q$, where cartesian product $Q \times \Sigma$ defines domain as single symbols only (all pairs (q,a) where $a \in \Sigma$). Multi-character "-1" requires $\delta: Q \times \Sigma^* \rightarrow Q$ (infinite domain), breaking finiteness since $Q \times \Sigma^*$ has infinite pairs.

Instead, we can add "minus-remembering" state after each original state, then from minus-state transitions with inputs 1,2,3 as negative values.

$Q = \{q_0, q_1, q_2, q_3, q_4, q_5, q_{0-}, q_{1-}, q_{2-}, q_{3-}, q_{4-}, q_{5-}\} \rightarrow$ these states remember if minus sign has been read from that state

$\Sigma = \{1, 2, 3, -\} \rightarrow$ minus sign added to alphabet

$S = q_0$

$F = \{q_0\}$

$\delta = \{$

$\delta(q_0, 1) = q_1 \quad \delta(q_0, 2) = q_2 \quad \delta(q_0, 3) = q_3$

$\delta(q_1, 1) = q_2 \quad \delta(q_1, 2) = q_3 \quad \delta(q_1, 3) = q_4$

$\delta(q_2, 1) = q_3 \quad \delta(q_2, 2) = q_4 \quad \delta(q_2, 3) = q_5$

$\delta(q_3, 1) = q_4 \quad \delta(q_3, 2) = q_5 \quad \delta(q_3, 3) = q_0$

$\delta(q_4, 1) = q_5 \quad \delta(q_4, 2) = q_0 \quad \delta(q_4, 3) = q_1$

$\delta(q_5, 1) = q_0 \quad \delta(q_5, 2) = q_1 \quad \delta(q_5, 3) = q_2$

$\rightarrow$ original transitions

$\delta(q_{0-}, -) = q_{0-}, \delta(q_{1-}, -) = q_{1-}, \delta(q_{2-}, -) = q_{2-}, \delta(q_{3-}, -) = q_{3-}, \delta(q_{4-}, -) = q_{4-}, \delta(q_{5-}, -) = q_{5-} \rightarrow$ minus read

$\delta(q_{0-}, 1) = q_5 \quad \delta(q_{0-}, 2) = q_4 \quad \delta(q_{0-}, 3) = q_3$

$\delta(q_{1-}, 1) = q_0 \quad \delta(q_{1-}, 2) = q_5 \quad \delta(q_{1-}, 3) = q_4$

$\delta(q_{2-}, 1) = q_1 \quad \delta(q_{2-}, 2) = q_0 \quad \delta(q_{2-}, 3) = q_5$

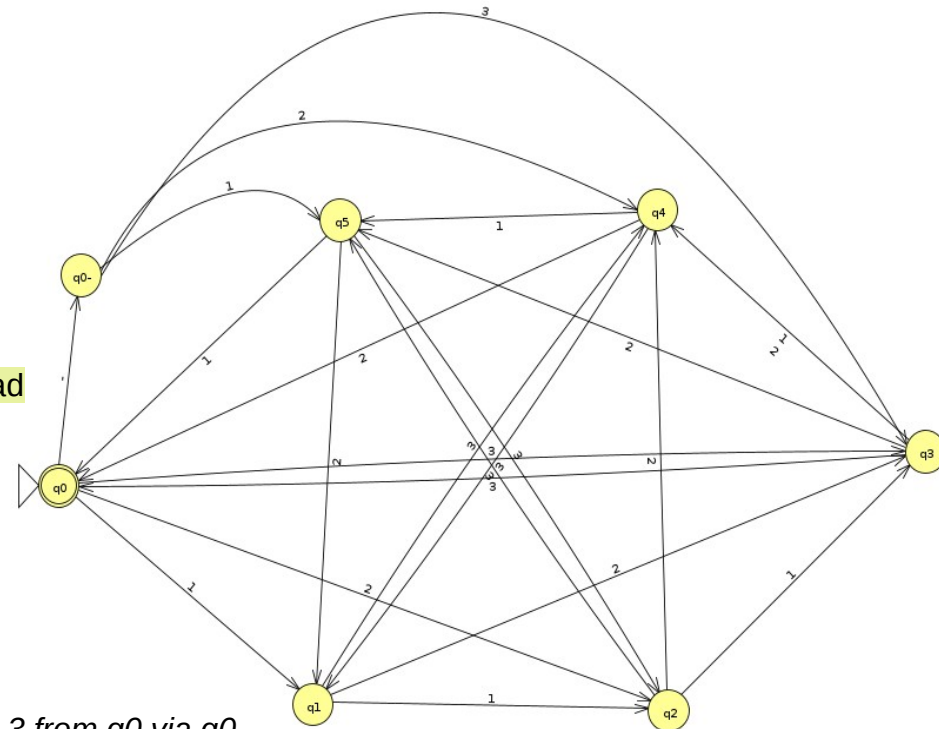
$\delta(q_{3-}, 1) = q_2 \quad \delta(q_{3-}, 2) = q_1 \quad \delta(q_{3-}, 3) = q_0$

$\delta(q_{4-}, 1) = q_3 \quad \delta(q_{4-}, 2) = q_2 \quad \delta(q_{4-}, 3) = q_1$

$\delta(q_{5-}, 1) = q_4 \quad \delta(q_{5-}, 2) = q_3 \quad \delta(q_{5-}, 3) = q_2$

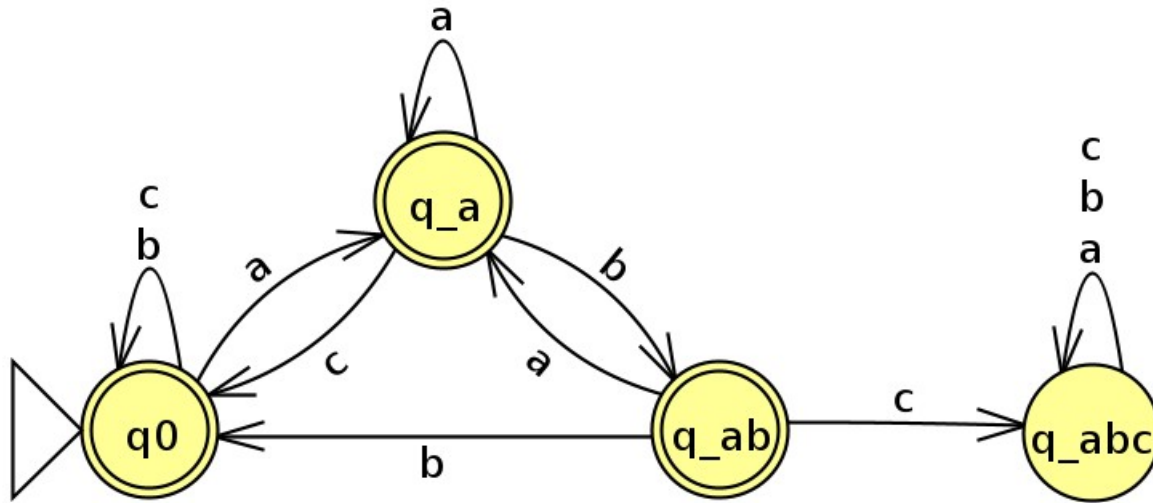
$\}$

$\rightarrow$ digit after minus sign



The image shows transitions with -1, -2, -3 from q0 via q0-

T3. Make an automaton ($\Sigma = \{a,b,c\}$) that does not accept any input containing the string abc as a part (but accepts all others).



T4. DFA is defined as follows:

$Q = \{q_0, q_1, q_2\}$

$\Sigma = \{0, 1\}$

$\delta = \{ \delta(q_0, 1) = q_0,$

$\delta(q_0, 0) = q_1,$

$\delta(q_1, 1) = q_0,$

$\delta(q_1, 0) = q_2,$

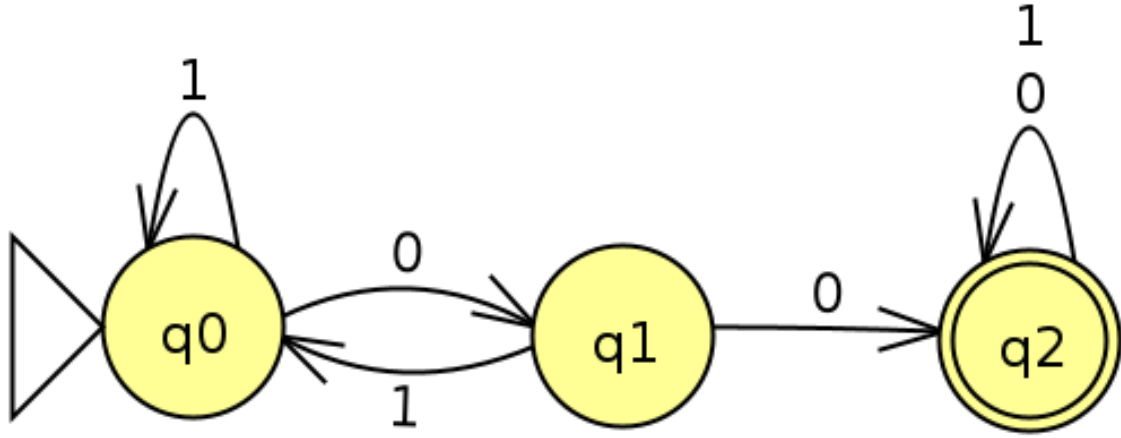
$\delta(q_2, 0) = q_2,$

$\delta(q_2, 1) = q_2,$

$\}$

$s = q_0$

$F = \{q_2\}$



Draw the automaton either by hand or with JFLAP.

What does the automaton do?

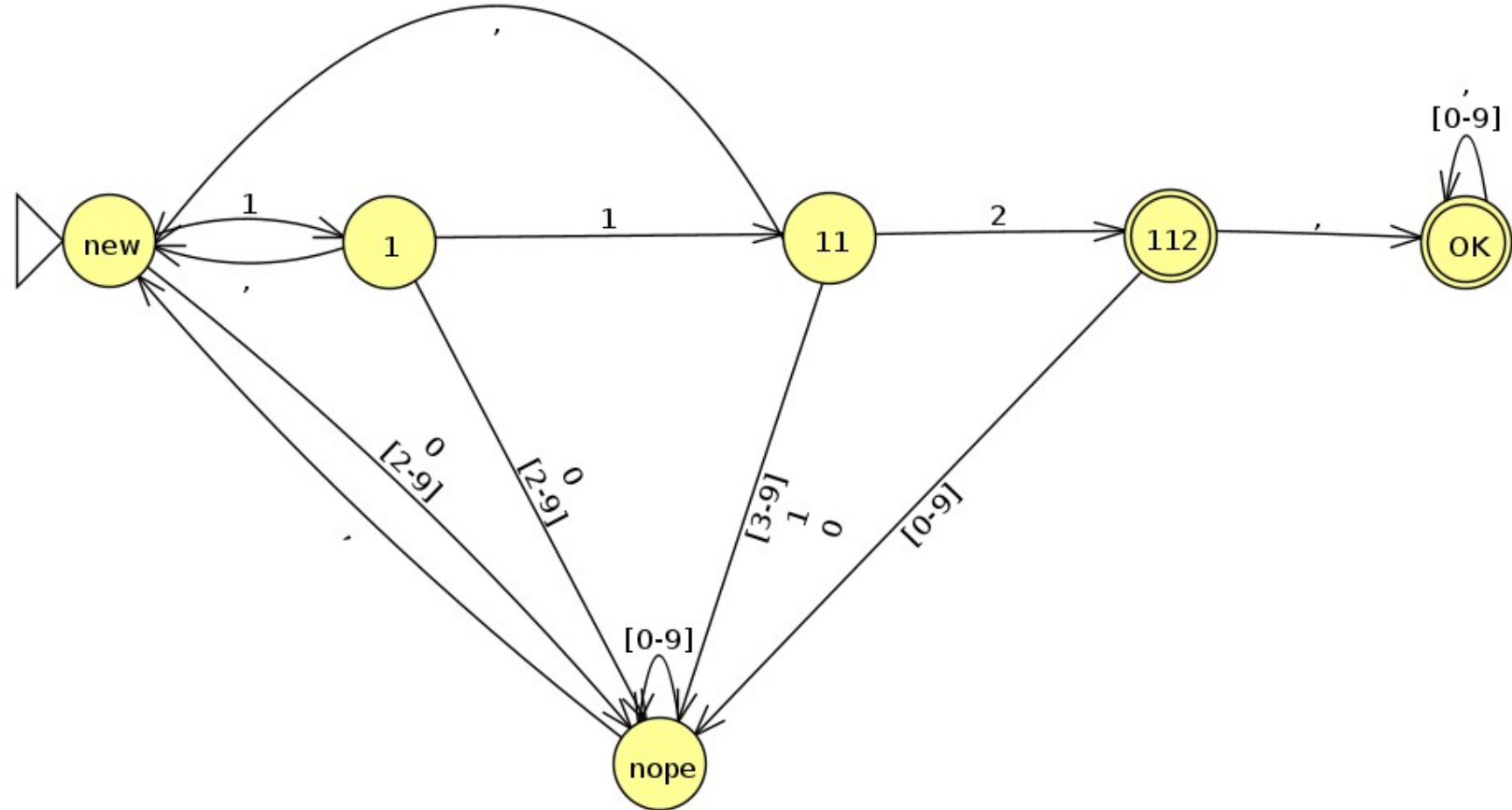
$L = \{w \in \{0,1\}^* \mid w \text{ contains substring "00"}\}$

Solve δ -transitions as a chain for 1101 and 01001, that is, expand $\delta(q_0, 1101)$ and $\delta(q_0, 01001)$

1 1 0 1
 $q_0 \rightarrow q_0 \rightarrow q_0 \rightarrow q_1 \rightarrow \mathbf{q_0}$

0 1 0 0 1
 $q_0 \rightarrow q_1 \rightarrow q_0 \rightarrow q_1 \rightarrow q_2 \rightarrow \mathbf{q_2}$

T5. Make a DFA that reads numbers separated by commas (,) from a list and accepts if the list contains the number 112, for example 2575,45777,9803,112,3567 would be accepted, but 295,430,1121,23,0 would not



T6. Let the alphabet be $\Sigma = \{a,b,c\}$. Construct an automaton that accepts exactly those strings in which

a) the number of the character a is odd

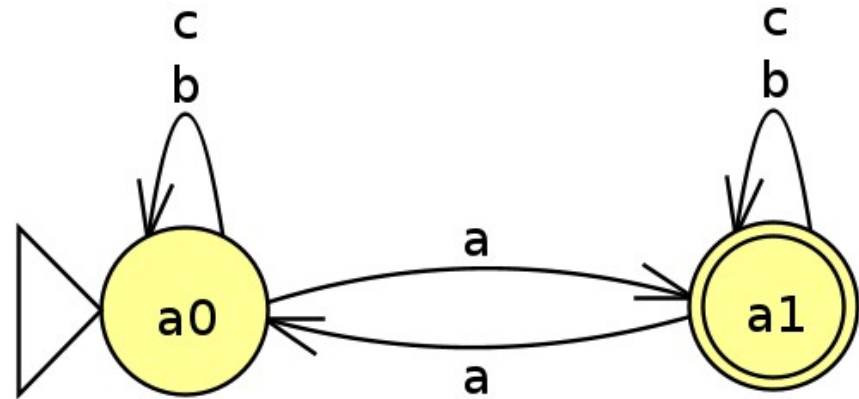
b) the number of the character a is odd or the number of the character b is even

c) either the number of the character a is odd or the number of the character b is even.

Justify in each case why your automaton accepts every string it should accept, but none of the strings it should not accept.

Rejects string when $|w|_a \bmod 2 = 0$

Accepts string when $|w|_a \bmod 2 = 1$



T6. Let the alphabet be $\Sigma = \{a,b,c\}$. Construct an automaton that accepts exactly those strings in which

a) the number of the character a is odd

b) the number of the character a is odd or the number of the character b is even

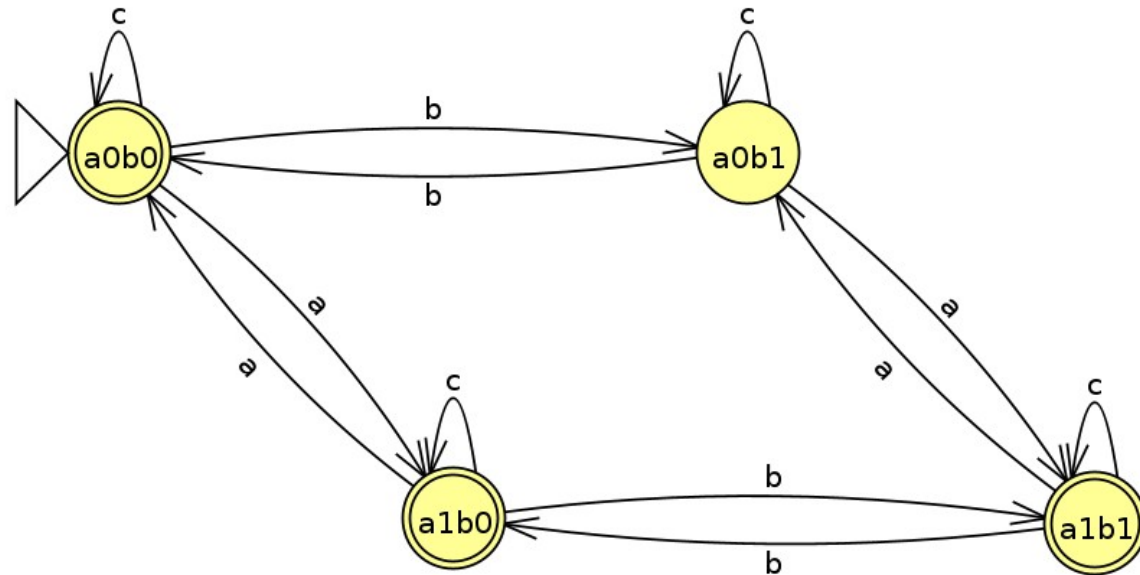
c) either the number of the character a is odd or the number of the character b is even.

Justify in each case why your automaton accepts every string it should accept, but none of the strings it should not accept.

$$|w|_a \bmod 2 = 1 \text{ OR } |w|_b \bmod 2 = 0$$

truth table

a	b	
0	0	1
0	1	0
1	0	1
1	1	1



T6. Let the alphabet be $\Sigma = \{a,b,c\}$. Construct an automaton that accepts exactly those strings in which

a) the number of the character a is odd

b) the number of the character a is odd or the number of the character b is even

c) either the number of the character a is odd or the number of the character b is even.

Justify in each case why your automaton accepts every string it should accept, but none of the strings it should not accept.

$$|w|_a \bmod 2 = 1 \text{ XOR } |w|_b \bmod 2 = 0$$

truth table

a	b	
0	0	1
0	1	0
1	0	0
1	1	1

